

Implementation Plan: Accessibility-Aware Maintenance Allocation for India's Road-Bridge-Culvert Network

Oct 8, 2026 · @navdhd

This is a 24-week plan to test whether allocating a fixed ₹ maintenance budget by loss of access to health and school services protects more people per rupee than condition-based ranking, with go/no-go at week 8.

1. Project definition

The project asks one question: does consequence-aware allocation beat the condition-based ranking that agencies actually use, and does the answer survive uncertainty in deterioration and repair effectiveness?

Research question. Does allocating a fixed ₹ budget by loss of access to essential services (health, schools) under monsoon-extreme hazards protect more people per rupee than condition-based and benefit-cost ranking, and how robust is that answer to uncertainty in deterioration and repair effectiveness?

Contributions to claim

1. An accessibility-loss consequence model for Indian road-bridge-culvert networks, built from open PMGSY, OpenStreetMap and IMD data.
2. A robust ₹ allocation method (expected loss plus tail risk, rolling horizon) with regret analysis over uncertain assumptions.
3. A reproducible benchmark: method validated on real labelled bridge histories (US NBI), deployed on Indian network data with stated assumptions.

Scope decisions

- **Asset classes:** roads, bridges and culverts only. No water, lighting or buildings.
- **Study regions:** one or two districts with different monsoon regimes, chosen after the week-1 feasibility check.
- **Currency and costs:** ₹, from state PWD Schedule of Rates where obtainable.
- **Human authority:** the system recommends; an engineer decides. No autonomous execution.

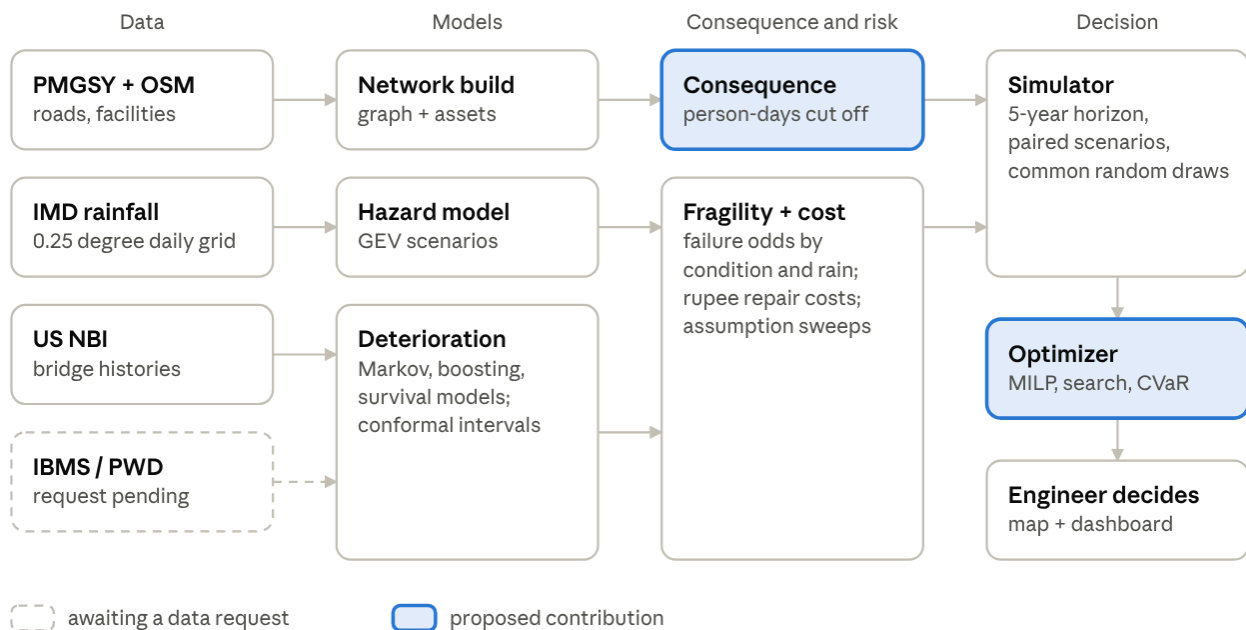
What you can and cannot claim

Claim	Status	Condition
Deterioration forecasts beat Markov baselines	Supported by real data	Validated on NBI, walk-forward by inspection year
Accessibility-aware allocation beats condition ranking	Supported in a simulator	Simulator validated component by component; conclusions shown under perturbed assumptions
Savings in ₹ or lives for a real district	Not claimable	Needs real Indian condition and failure records
Multimodal prediction beats single-source	Only if Indian labels are obtained	Otherwise drop this research question

2. Architecture

Open data feeds four models; a simulator and an optimizer turn them into a plan that an engineer approves.

Open data flows through four models into one allocation decision



system architecture · 4 data sources, 3 model stages, 1 decision

The two highlighted boxes are the project's own contributions: the accessibility-loss consequence and the robust optimizer. The dashed source is the data you must request; everything else is open.

3. Data plan

Three of four layers have open Indian data; the fourth, asset condition and failure, is the bottleneck and needs a request that should start in week 1.

Layer	Source	Access	Used for	Status
Rural road network, habitations, facilities	PMGSY Rural Connectivity datasets	Open, Government Open Data License	Graph, population served, health and school locations	Checked; quality varies by state
Road network, structure tags	OpenStreetMap (Geofabrik India extract, OSMnx)	Open, ODbL attribution	Highways, bridge and culvert tags, graph connectivity	To verify tag coverage per district
Rainfall hazard	IMD gridded rainfall, 0.25°, 1901–present via imdlib	Open download from IMD Pune	Extreme-value fits, return-period scenarios	Checked
Bridge and culvert condition (India)	IBMS (MoRTH) or state PWD records	No public download found	Calibration of deterioration; real condition ratings	Request needed
Bridge condition histories (US)	FHWA National Bridge Inventory , annual files	Open	Method validation with real inspection histories	Checked
Unit costs	State PWD Schedule of Rates	Usually on PWD or e-procurement sites	₹ repair costs per intervention	To verify
Population (fallback)	Gridded population raster	Open	Habitation population if the PMGSY layer lacks it	To verify

Data rules

- Raw files are immutable under `data/raw/` and versioned with DVC; every derived table records its source, date and licence.

- Attribution is kept for GODL (PMGSY) and ODbL (OpenStreetMap); IMD terms are checked before any redistribution.
- Every table in the paper states whether its numbers come from real data, a fitted model or an assumption.

Assumptions register (a living table)

Assumption	Why needed	How varied
Fragility: failure probability given condition and rainfall intensity	No Indian failure labels	Parametric family with literature-set priors; sweep slope and threshold
Deterioration acceleration for Indian conditions vs NBI	Climate, load and maintenance differ	Multiplier range, reported as a sensitivity axis
Repair effectiveness (condition reset, risk reduction)	Not observable in historical data	Range per intervention type; regret reported
Outage duration after failure	No repair-time records	Distribution by structure type; sweep mean
Facility weights (health vs school)	Policy choice	Pareto front and rank-stability tests

4. Stack and repository

The stack is Python for everything analytical and a thin FastAPI and React layer on top; nothing here needs a GPU.

Concern	Choice
Language and env	Python 3.11, <code>uv</code> or <code>conda</code> , <code>ruff</code> , <code>pytest</code>
Geospatial	GeoPandas, Shapely, pyproj, rasterio, xarray, OSMnx
Graph	python-igraph (fast shortest paths), NetworkX for prototypes
Rainfall	<code>imdlib</code> (Python) for IMD gridded data
Extreme values	SciPy <code>genextreme</code> (GEV), L-moments check

Concern	Choice
Deterioration	scikit-learn, LightGBM, lifelines (survival), hand-written Markov estimator
Uncertainty	Conformal prediction (MAPIE or your own, to show you understand it)
Optimization	OR-Tools (SCIP or CP-SAT) or Pyomo with HiGHS
Storage	Parquet and DuckDB for analysis; PostGIS only if the dashboard needs it
Tracking	MLflow for experiments, DVC for data versions
API and UI	FastAPI; React + Vite with MapLibre GL
Packaging	Docker Compose, GitHub Actions for tests

Repository layout

```

road-resilience-allocation/
├─ data/                raw/ interim/ processed/    (DVC-tracked)
├─ src/rra/
│   ├─ ingest/          pmgsy.py osm.py imd.py nbi.py ibms.py
│   ├─ network/         build.py snap.py attach_assets.py
│   ├─ hazard/          gev_fit.py scenarios.py
│   ├─ consequence/     accessibility.py population.py
│   ├─ deterioration/   markov.py gbm.py survival.py conformal.py
│   ├─ fragility/       fragility.py
│   ├─ cost/            sor.py interventions.py
│   ├─ simulator/       state.py step.py policies.py
│   ├─ optimize/        milp.py local_search.py cvar.py rolling.py
│   ├─ evaluate/        metrics.py stats.py ablation.py regret.py
│   └─ workflow/        dag.py report.py narrate.py
├─ experiments/         e01_deterioration/ ... e08_ablation/
├─ api/                 FastAPI app
├─ web/                 React + MapLibre dashboard
├─ notebooks/           exploration only, never the source of results
├─ tests/               unit, property, regression
├─ docs/                assumptions.md, data_cards/, paper/
├─ Dockerfile           docker-compose.yml dvc.yaml mlflow/
└─ README.md

```

Engineering rules

- Every experiment is a script with a config file and a fixed seed; notebooks never produce a number that goes in the paper.

- Common random numbers: all policies are evaluated on the same scenario draws, so differences are paired.
- Tests use small hand-made networks where the right answer is known (see section 5.3).

5. Module specifications

Seven modules feed each other in a fixed order: network, hazard, consequence, deterioration, fragility and cost, simulator, optimizer. Each has inputs, method, outputs and tests.

5.1 Network build

Goal: a routable graph per study district with habitations, facilities and structures attached.

1. Load district boundary; clip PMGSY road centerlines, habitations and facility points.
2. Pull OSM highways for the same area; prefer PMGSY geometry for rural roads and OSM for state and national highways.
3. Snap and planarize (tolerance is a parameter, start near 25 m), then build an igraph graph with length, road class and free-flow speed by class (speeds are an assumption in the register).
4. Attach each habitation and facility to its nearest node within a distance cap; log those beyond it.
5. Attach structures from OSM `bridge` and `culvert` tags, and from IBMS records if you obtain them; split edges at structure locations.
6. QA report: connected components, share of habitations connected, orphan facilities, tag coverage.

Output: `network.parquet`, `graph.graphml`, `qa_report.md`. **Exit test:** more than 95% of habitations reach at least one health facility in the intact network, or the district is rejected.

5.2 Hazard model

Goal: rainfall scenarios with known return periods for every road segment.

- Download IMD rainfall with `imdlib`; compute annual maxima of 1-day and 5-day rainfall per 0.25° cell.
- Fit a GEV per cell, pooling neighbouring cells where records are short; check fit with a goodness-of-fit test and QQ plots.
- Report return levels for 2, 5, 10, 25, 50 and 100 years.

- **Scenarios:** (a) resample historical monsoon years as spatial fields, which keeps real spatial correlation; (b) stress events scaled to a chosen return level.
- Optional: test for a trend in the GEV location parameter and report it as a sensitivity, not a forecast.

Output: `hazard_scenarios.zarr` with N scenarios by cell. **Tests:** return levels increase with period; fitted tails stay finite.

5.3 Accessibility-loss consequence

Goal: the person-days of access lost if an asset, or a set of assets, fails. This replaces traffic-volume criticality.

For a failure set F , travel time to the nearest facility of type k changes for each habitation h . With population P_h , facility weight w_k and a cap T_k on acceptable travel time:

$$L(F) = \sum_{h \in H} \sum_{k \in K} P_h w_k \left[\phi(\tau_k^F(h)) - \phi(\tau_k^0(h)) \right], \quad \phi(\tau) = \min(1, \tau/T_k)$$

Here τ^0 is travel time on the intact network and τ^F with all assets in F removed; an unreachable facility counts as the cap. Societal cost of an event is $C(F) = D \times L(F)$, with D the outage duration in days.

- **Non-additivity is the point.** Two parallel routes each cost zero alone and a lot together, so $L(F)$ is computed on the whole set, never summed per asset.
- **Speed-up:** run multi-source Dijkstra from facilities on the reversed graph, and only recompute for habitations whose shortest-path tree uses a failed edge.
- **Population:** use the PMGSY habitation population if present, otherwise a gridded population raster allocated to the nearest habitation.
- **Weights w_k and caps T_k** are policy choices; report results over a grid and show rank stability.

Tests (hand-built toy graphs): a single bridge on the only route gives loss equal to cut-off population; parallel routes give zero individually and positive jointly; loss is non-negative and monotone as more assets are added to F .

5.4 Deterioration model

Goal: a distribution over future condition for each structure, with calibrated uncertainty.

Validated on NBI, then transferred.

- **Target:** overall condition from the deck, superstructure, substructure and culvert ratings (check the coding guide for the exact rule), collapsed to four states: Good, Fair, Poor, Severe. Second target: time to Poor.
- **Models:** (M0) age-class Markov chain, the agency-style baseline; (M1) gradient boosting for next-year state; (M2) discrete-time survival hazard for time to Poor.
- **Features from NBI itself:** age, structure type and material, span and width, traffic, state or region. Add weather only if time allows.
- **Split by inspection year,** never random: train on early years, validate on the middle, test on the latest; plus a held-out-state test. Adjust year ranges to what you download.
- **Metrics:** log-loss and ranked probability score for state forecasts, concordance index for time to Poor, calibration of P(Poor within 5 years), and conformal coverage by group.
- **Uncertainty:** split-conformal intervals on time to Poor (class-conditional by structure type), plus bootstrap uncertainty on Markov transition rows.

Transfer to India. Map IBMS rating definitions onto the four states once you have them. Apply an acceleration multiplier θ to hazard rates, justified from published Indian studies and swept as a sensitivity axis. If IBMS or PWD data arrive, recalibrate on a held-out slice and report coverage before and after.

5.5 Fragility, outage and cost

- **Fragility:** probability that asset a fails in a monsoon event, given condition c and rainfall ratio R to its local reference level:

$$\Pr(\text{fail}_a \mid c, R) = \sigma(\beta_0 + \beta_1 g(c) + \beta_2 \log(R/R^{\text{ref}}_a))$$

Parameters per structure type (culvert, minor bridge, major bridge, road segment) are assumptions with literature-informed ranges; you vary them rather than claim them.

- **Outage duration D :** a distribution by structure type, also in the assumptions register.
- **Interventions:** none, minor repair, major repair or replacement. Each has a ₹ cost from the state Schedule of Rates, a labour requirement, and an effect on condition and on fragility.
- **Emergency premium:** a failed asset costs more to restore than a planned repair; the multiplier is a swept parameter.

5.6 Policy simulator

Goal: compare policies on the same futures.

One simulated year does the following: apply the policy's interventions within budget; deteriorate each structure using the fitted model; draw a monsoon scenario; draw failures with common random numbers; compute $L(F)$, outage cost and emergency repair cost; log everything.

- Run 5-year horizons over 500 to 1,000 scenarios; reuse the same random draws for every policy so comparisons are paired.
- **Validation, component by component:** deterioration against held-out NBI years; GEV fit against observed rainfall; accessibility loss against toy cases. No component is validated by the simulator's own output.
- **Output:** per policy, distributions of person-days of access lost and ₹ spent, plus a tail metric (CVaR).

5.7 Optimizer

Goal: choose interventions that minimize expected societal loss plus tail risk under a budget and crew capacity.

With common random numbers, whether asset a fails in scenario ω under option j is a fixed binary f , so the additive version is a linear program with binary decisions x :

$$\min_x \frac{1}{|\Omega|} \sum_{\omega \in \Omega} \ell_{\omega}(x) + \lambda \text{Big}(\eta + \frac{1}{\alpha} (1 - \sum_{\omega \in \Omega} z_{\omega} \text{Big}))$$

$$\text{s.t.} \quad z_{\omega} \geq \ell_{\omega}(x) - \eta, \quad z_{\omega} \geq 0, \quad \sum_{a,j} c_{aj} x_{aj} \leq B, \quad \sum_{a,j} u_{aj} x_{aj} \leq W, \quad \sum_j x_{aj} \leq 1$$

Here ℓ is the scenario loss, c the ₹ cost, u the labour need, B the budget and W the crew capacity.

- **Additive MILP (baseline):** $\ell_{\omega}(x) = \text{sum of } C_a \text{ times the failure indicator}$. Solves exactly with OR-Tools or HiGHS.
- **Network-aware search:** because real loss is non-additive, start from the MILP solution and improve with swap-based local search on the sample-average objective, evaluating $L(F)$ per scenario on sets. Tune on 200 scenarios; report on 1,000 fresh ones.
- **Rolling horizon:** re-solve every year with updated conditions; compare with one-shot 5-year plans.
- **Quality check:** on small instances, report the gap between local search and the exact solution of a linearized model. No optimality guarantee is claimed for the full problem.
- **Constraints to include:** annual budget, crew-days, a minimum protected access for health facilities (as a stated constraint), and optional fairness across blocks.

Property tests: more budget never raises optimal loss; zero budget reproduces the do-nothing policy; results are reproducible under a fixed seed.

6. Evaluation protocol

The primary metric is person-days of access to health and school services lost per ₹ crore spent, over a 5-year horizon, paired across policies on identical scenarios.

Policies compared

Policy	What it does	Why it is included
Do nothing	No interventions	Floor
Random	Random structures within budget	Sanity check
Worst-first	Lowest condition first	Common agency practice
Benefit-cost ranking	Risk reduction per ₹, additive, no network	Standard decision-support method
Criticality-weighted	Condition times a socio-economic rating proxy	Stands in for the IBMS rating idea
Siloed budgets	Fixed budget split per asset class, then best within class	Tests the unified-allocation claim
Additive MILP	Exact on the additive model	Optimization without network effects
Network-aware search	Local search on set-based loss	Proposed method
Network-aware + CVaR + rolling	Proposed method with tail risk and re-planning	Full system

Experiments

ID	Question	Comparison	Primary measure
E1	Do ML models beat agency-style deterioration models?	M0 vs M1 vs M2 on NBI	Log-loss, concordance, calibration
E2	Do conformal intervals hold?	Static vs recalibrated, by group	Coverage vs target

ID	Question	Comparison	Primary measure
E3	Does accessibility-aware allocation beat condition ranking?	Worst-first vs benefit-cost vs proposed	Person-days lost per ₹ crore
E4	Do network effects matter?	Additive MILP vs network-aware search	Loss at equal budget
E5	Does unified beat siloed?	Siloed vs unified, across budgets	Loss at equal total budget
E6	Does uncertainty help?	Point vs interval-aware vs CVaR	Mean and CVaR loss
E7	Does re-planning help?	One-shot vs rolling horizon, under extreme-rain scenarios	Loss and budget use
E8	How robust are conclusions?	Vary assumptions in the register	Win rate and regret

Ablation: remove one component at a time from the full system (network effects, uncertainty, CVaR, rolling horizon, rainfall hazard, facility weighting) and report the marginal change in the primary metric.

Statistics

- Same scenario draws for every policy; report paired differences with bootstrap confidence intervals, resampling scenarios and, for cross-region claims, resampling regions.
- At least 5 seeds for stochastic searches; report medians and ranges.
- Pre-register the cost and weight settings before running E3 to E8; report any later change.
- Tune on one scenario set, report on a fresh one.

Robustness and regret. For each policy, report the fraction of sampled assumption sets in which it wins, and its regret: the loss gap to the best policy under that assumption set. The headline claim is a policy that is near-best across the whole range, not one that wins under a single setting.

Hypotheses

- **H1:** Accessibility-aware allocation protects more access per ₹ than condition-based ranking across rainfall scenarios.

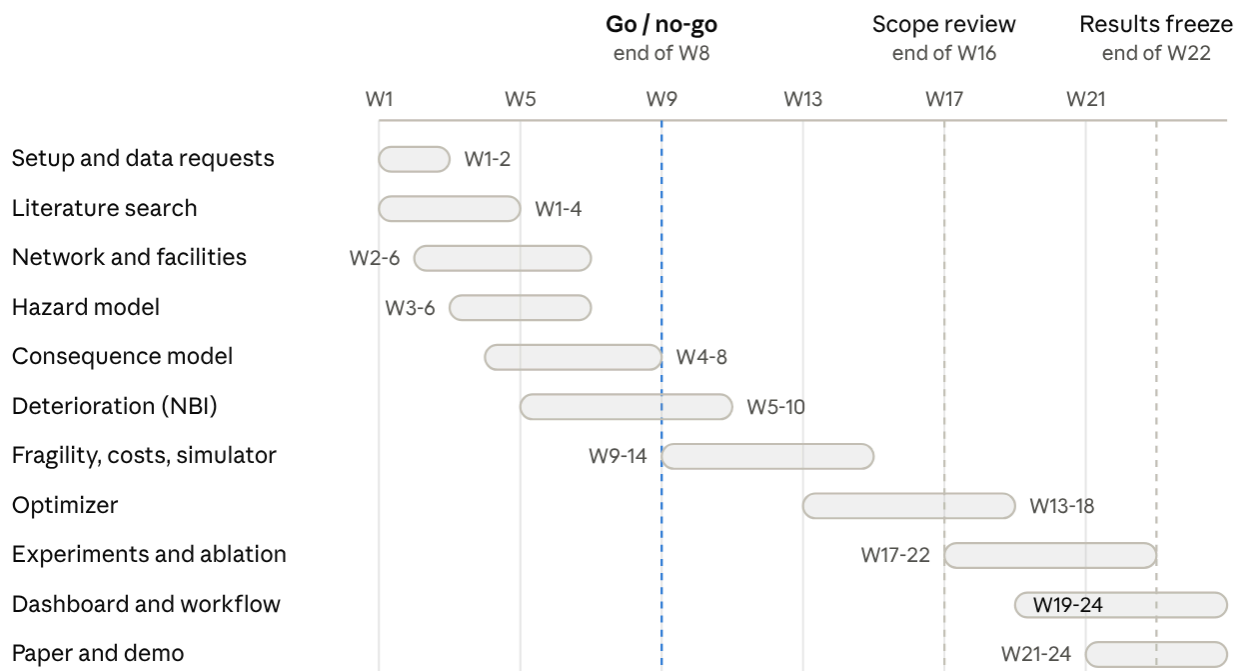
- **H2:** The advantage survives perturbation of deterioration and repair-effectiveness assumptions (low regret).
- **H3:** Interval- and CVaR-aware allocation lowers tail loss compared with point-estimate allocation.
- **H4:** Unified allocation across asset classes beats per-class budgets at tight budgets.
- **H5:** Conclusions hold in a second district with a different monsoon regime.

Expect H4 to look weak at generous budgets. Report where each hypothesis fails; those results are part of the contribution.

7. Schedule

The plan runs 24 weeks in six four-week sprints, with a go/no-go at the end of week 8, a scope review at week 16 and a results freeze at week 22.

Week 8 decides whether the plan proceeds as written



24-week schedule · 11 workstreams, 3 gates

Start week 1 when you choose; every date below is relative to it.

Sprint	Weeks	Deliverables	Exit criterion
1. Foundation and feasibility	1-4	Repository and environment; data requests sent; one district graph with facilities; IMD extraction; accessibility-loss prototype checked on toy graphs; literature table started	One district's consequence map reproduces from raw data with one command
2. Deterioration and hazard	5-8	NBI ETL and cross-year joins; Markov, boosting and survival baselines; walk-forward validation (E1); GEV fits and rainfall scenarios; first mapping of Indian ratings	Go/no-go: district passes network QA; E1 runs with honest baselines; data-fallback decision made
3. Uncertainty and simulator	9-12	Conformal intervals and coverage (E2); fragility, outage and ₹ cost tables; simulator with baseline policies; paired-scenario harness	Do-nothing, worst-first and benefit-cost policies run end to end with confidence intervals
4. Optimizer	13-16	Additive MILP; network-aware local search; CVaR; rolling horizon; first results for E3 to E5	Scope review: the method beats a baseline in some regime, or the framing shifts to "when does allocation by access help"
5. Experiments and robustness	17-20	E3 to E8, ablation, second district, regret analysis, statistics	Every result regenerates from a config file; draft results table
6. Product and write-up	21-24	Results freeze at week 22; dashboard and API; orchestrator with narration check; Docker; paper draft; demo rehearsal	One-command reproduction works; demo runs offline

Cut order if behind (the last item is never cut):

1. Orchestrator polish and LLM narration.
2. Dashboard views beyond the map and the budget slider.

3. The second district; keep H5 as future work.
4. The survival model M2.
5. Rolling-horizon re-planning (keep one-shot).
6. *Never cut*: network-aware consequence, paired evaluation, the assumptions register, and honest baselines.

8. Product layer: orchestrator, API and dashboard

The workflow is a deterministic pipeline of tested functions; a language model only turns finished numbers into readable text, which keeps every figure traceable.

Orchestrator

- A small DAG runner in `workflow/dag.py` : each node is a pure function with typed inputs and outputs, logged to MLflow.
- A request such as "what should we repair this year with ₹ X crore?" maps to a fixed plan: load district, load scenarios, forecast condition, compute consequence, optimize, report.
- **LLM role**: narration only. It receives a JSON of results and writes the summary. A check rejects any number in the text that is not in the JSON.
- **Human authority**: each recommendation lists its assumptions, its uncertainty interval, and a one-click "inspect first" flag for structures where the prediction is uncertain.

Evaluating the workflow (small, honest)

- 30 to 50 test requests with expected plans: check correct tool sequence and failure handling.
- Numeric faithfulness: share of figures in generated text that match the result JSON.
- Drop any claim comparing "multi-agent vs single-agent" unless you can run it fairly.

API (FastAPI)

Endpoint	Returns
GET <code>/districts</code>	Available study districts and data status
GET <code>/districts/{id}/assets</code>	Structures with condition forecast and interval
GET <code>/districts/{id}/consequence</code>	Accessibility loss per asset, with scenario range
POST <code>/optimize</code>	Recommended interventions for a budget, horizon and risk setting
POST <code>/simulate</code>	Policy comparison on a scenario set

Endpoint	Returns
GET /assumptions	The assumptions register in force

Dashboard (React + MapLibre)

1. **Map view:** roads and structures coloured by accessibility loss; habitations cut off in a chosen failure scenario.
2. **Budget slider:** side-by-side plans (condition-based vs proposed) with expected and tail loss.
3. **Asset card:** why this asset ranks here, with condition forecast, interval, consequence and cost.
4. **Rainfall scenario switcher:** 10-, 25- and 100-year events and the re-planned recommendation.
5. **Assumptions panel:** change a parameter and see rankings move.

Quality gates: unit and property tests in CI, a data-card per dataset, MLflow run IDs on every dashboard number, and a README that reproduces the headline result from raw data in one command.

9. Risks and mitigations

The biggest risks are data access and the temptation to present simulator results as field evidence; both are handled by planning for the no-data case from day one.

Risk	Likelihood	Impact	Mitigation	Trigger
IBMS or PWD condition data not obtained	High	High	Frame as methodology with NBI-validated deterioration and assumption sweeps; keep sending requests	No reply by week 6
OSM bridge and culvert tags too sparse	Medium	High	Compare tag counts to expected structure density; fall back to inferring crossings where roads meet water bodies, or use PMGSY structures if present	Coverage low in week 2

Risk	Likelihood	Impact	Mitigation	Trigger
PMGSY quality uneven across states	Medium	Medium	Choose districts with complete road and habitation layers; document QA	QA exit test fails
Habitation population missing	Medium	Medium	Gridded population fallback; sensitivity to allocation method	Week 2
Simulator conclusions are circular	Medium	High	Validate each component against real data separately; report robustness over the assumptions register	Reviewer objection
Consequence weights look arbitrary	High	Medium	Pareto fronts, rank-stability tests, pre-registered defaults	Always present
Compute cost of per-asset network recomputation	Medium	Medium	Recompute only affected habitations; cache; limit scenarios during search	Runs exceed an hour
Local search gives poor solutions	Medium	Medium	Seed from MILP; compare with exact on small cases; multiple restarts	Gap above 5% on toys
Scope creep into water, sensors, RL or agents	High	High	Keep these in the stretch list; review scope at weeks 8 and 16	New module proposed
LLM text contains wrong numbers	Medium	Medium	Narration only, with automatic number-matching check	Check fails
US-to-India transfer questioned	High	Medium	Say it plainly; validate method on NBI, apply to India with sensitivity axes	Viva

Risk	Likelihood	Impact	Mitigation	Trigger
Time overrun	Medium	High	Fixed-length sprints; two-week buffer in month 6; cut order is fixed (section 7)	Week 20 behind plan

10. Week-1 checklist

Week 1 has one goal: prove you can build a routable district graph with facilities and compute one accessibility-loss number, while the data requests go out.

- ☐ Create the repository, environment and folder layout from section 4; commit a README with the research question.
- ☐ Send a data request for bridge and culvert condition records to a state PWD and to MoRTH (RTI as a backup); keep a log of dates and replies.
- ☐ Download the PMGSY datasets for two candidate districts; note state-level quality.
- ☐ Download an OSM extract and count `bridge` and `culvert` tags in each candidate district.
- ☐ Install `imdlib`; download one rainfall year and extract a district time series.
- ☐ Build the road graph for one district and attach habitations and health facilities.
- ☐ Compute accessibility loss for removing one edge; check it against a hand calculation.
- ☐ Download 3 NBI years for 2 states and join structures across years.
- ☐ Write the first version of the assumptions register (section 3) and commit it.
- ☐ Pick the district pair, or write down what is blocking the choice.

11. Paper outline and demo script

The paper's argument is simple: condition-based ranking misallocates scarce budget because it ignores who loses access, and a network-aware, uncertainty-aware method does better across plausible assumptions.

Paper outline

1. **Abstract and introduction:** the maintenance-allocation problem in India, and why consequence is missing from condition-based ranking.
2. **Related work:** bridge management systems and deterioration models; accessibility and network criticality; conformal prediction; robust and CVaR optimization; infrastructure resilience under hazards.
3. **Problem formulation:** network, hazard, accessibility loss, interventions, budget.

4. **Methods:** sections 5.1 to 5.7.
5. **Data and validation setup:** what is real, what is fitted, what is assumed.
6. **Results:** E1 to E8 with ablation and regret.
7. **Limitations:** no Indian failure labels unless obtained, US-to-India transfer, simulator-based policy value, population and speed assumptions.
8. **Conclusion and future work:** real condition data, water and drainage assets, learned policies.

Run a proper literature search in month 1 and write the related-work table early; do not claim novelty until it is done.

Demo script (8 minutes)

1. **Problem (1 min):** a district map; a worst-first plan repairs the lowest-rated bridge.
2. **Consequence (2 min):** switch to accessibility loss; one "unremarkable" culvert would cut off a block from its only health centre.
3. **Uncertainty (1 min):** its condition forecast has a wide interval, so the system flags it for inspection first.
4. **Budget (2 min):** move the ₹ slider; show proposed plan against worst-first on expected and tail loss.
5. **Storm (1 min):** switch to a 25-year rainfall scenario; the plan re-optimizes.
6. **Honesty (1 min):** open the assumptions panel; show that the ranking holds across the range.

12. Sources

These facts came from search results on 8 Oct 2026, not from reading each page in full; open them before citing anything in the paper.

- [PMGSY Rural Connectivity open data](#): road centerlines, habitations, facility points, Government Open Data License.
- [OpenStreetMap wiki: PMGSY data import](#): dataset contents and the note that quality varies by state.
- [IMDLIB documentation](#): downloading and reading IMD gridded rainfall and temperature in Python.
- [FHWA National Bridge Inventory downloads](#): annual NBI files and record format.
- [Indian Bridge Management System overview](#): about 1.72 lakh structures inventoried, mostly culverts.
- [MoRTH nationwide bridge survey, July 2026](#): new IBMS mobile app and agency dashboard.

Figures and methods taken from memory, to verify: the exact NBI condition-state rule, OSMnx and Geofabrik usage, state PWD Schedule of Rates availability, and the conformal and CVaR formulations.